Rank Access and Closure Depth in the Relation-Layer Cohomology of Token Arenas

A refuted conjecture, a refuted mechanism, and two species of invariant harmonic residue

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Abstract

A K_{disj}-type relation layer over a token arena assigns cohomology to a graph by declaring which closed loops count as relations. We settle an open problem about such layers and then identify what actually governs them. First, the conjecture that two vertex orbits force $\varepsilon_{harm}(K_4) = 0$ is refuted by an exact 12-vertex counterexample with $|Aut| = 24$, reproduced by two implementations whose constructions of the invariant cochain space share no logic. Second, we localize the failure: it is not a shortage of cells. On the witness's two-dimensional residue, all 1083 arena 4-cycles supply rank 1, while 20 arena 3-cycles of an excluded shape supply rank 2, and the two nonzero square families have the *same* image, a single line. Third, we refute the natural mechanism for this — that the residue is an odd-cycle blind spot — by exhibiting bipartite arenas, containing no odd cycle of any length, that still carry residue. The operative property is omission from the declared family, not parity. Fourth, we prove that $\varepsilon_{harm}(K_{3,4}) > 0$ on an infinite family — every cycle graph with $n \geq 5$ and $2 \leq K \leq n-1$, under the trivial group and the rotation subgroup, and zero under the full dihedral group — answering a question the odd-cycle reading had posed. Fifth, we define closure depth $L_{closure}$, prove it exceeds 4 for every member of that family, and measure it as exactly n for C_n and 4 for the witness, so no short vocabulary closes the family. This separates two populated regimes: residue a longer reader dissolves, and residue carried by configuration-space topology. Finally we identify the second kind. At trivial group the framework's own quantity is simplicial H^1 over $\mathbb{Q}$, and the declared cells are the 2-skeleton of Abrams' discretized configuration space; the cell counts

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and Euler characteristic are verified against closed form, Abrams' subdivision hypothesis for C_n reduces to $n \geq K + 1$, the measured rank is 1 on all 33 admissible rows and 0 on all 8 excluded boundary rows, and the equivariant degree is solved exactly: +1 for rotations, −1 for reflections. All arithmetic is exact rational.

Keywords: graph configuration space, graph braid group, token graph, discrete Morse theory, equivariant cohomology, cube complex, relation layer, closure depth

MSC 2020: 55R80, 05C10, 20F36, 55N91, 05C85

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1. Introduction

A graph does not come with a cohomology. One is obtained by declaring a relation layer: a set of closed loops in some derived space that are stipulated to bound. Change the declaration and the cohomology changes while the graph does not. This paper is about what that dependence actually is, measured rather than described.

The derived space here is a *token arena*. Fix a finite simple graph X, an integer $K \geq 1$, and a group $G \leq \mathrm{Aut}(X)$. The arena $A_K(X)$ has the K-element subsets of $V(X)$ as vertices, with an edge whenever two subsets differ by moving one token along one edge of X. A relation layer declares a family of arena loops as 2-cells; the G-invariant cochain complex of the resulting 2-complex has a first cohomology whose dimension we write $\varepsilon_{\mathrm{harm}}$. The quantity is relative to the declaration, and that is the point.

Two declarations are natural and were already in use. K_{disj} admits only *commuting squares*: two token moves along base edges with disjoint closed supports, which therefore commute. K_4 admits every 4-cycle of the arena. Since $K_{\mathrm{disj}} \subseteq K_4$, monotonicity gives $\varepsilon_{\mathrm{harm}}(K_4) \leq \varepsilon_{\mathrm{harm}}(K_{\mathrm{disj}})$, and the interesting question is when the larger family closes what the smaller one leaves open.

1.1 What this paper settles

The conjecture under test asserted that a substrate with exactly two vertex orbits forces $\varepsilon_{\mathrm{harm}}(K_4) = 0$. It is false. Section 3 exhibits a 12-vertex witness satisfying the hypothesis with $\varepsilon_{\mathrm{harm}}(K_4) = 1$, certified in exact rational arithmetic and reproduced independently.

Refuting a conjecture is the cheap part. The substance is in sections 4 through 8, which answer: *what is the residue, and what reads it?* Three successive mechanisms were proposed during the investigation and two were killed by their own predictions. What survives is a statement about rank, a statement about closure depth, and an identification of one residue species against a cited theorem whose hypothesis we verify and whose boundary we test.

1.2 Summary of results

	§	Statement	Status
A	§3	The two-orbit conjecture is false: a 12-vertex, |Aut| = 24, two-orbit witness has εharm(K4) = 1.	refuted by exact witness
B	§4	On the witness's 2-dimensional residue, 1083 arena 4-cycles have rank 1; 20 arena 3-cycles have rank 2. The spectator and rotation square families share one image line.	computed

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	§	Statement	Status
C	§5	The odd-cycle mechanism is false: bipartite arenas with no odd cycle of any length still carry residue.	refuted by witness family
D	§6	εharm($K_{3,4}$) > 0 on an infinite family: Cn for every n ≥ 5 and 2 ≤ K ≤ n−1, under G = 1 and G = Zn; zero under D2n.	PROVED (Thm 6.3), given [1]
E	§7	Lclosure = 4 for the witness; = n for Cn on every tested row. Lclosure > 4 for all n ≥ 5 is proved.	computed; partly proved
F	§8	At trivial G the computed quantity is dim H^1(UDK(X); ℚ); Abrams' hypothesis for Cn is n ≥ K+1; measured 1 on 33 admissible rows, 0 on 8 boundary rows; equivariant degree +1 / −1.	cited theorem, hypothesis verified, prediction confirmed

Table 1. The results, with the epistemic status of each. No entry is a theorem proved here; A and C are refutations by explicit witness, B, D and E are exact finite computations, and F is a cited equivalence whose hypothesis is checked and whose prediction is confirmed across its admissible range and contradicted at its first excluded row.

1.3 Background, and what is new here

The toolkit is established and no part of it is claimed here. Unordered K-token states on a graph, collision-free token moves, the cubical model in which independent moves supply cells, H^1 measuring exchange and traversal loops, the homotopy type $\mathrm{UConf}_K(S^1) \simeq S^1$, and the fact that a rotation preserves while a reflection reverses an orientation class on a circle — all of this is standard graph-configuration-space and graph-braid-group topology [1, 3, 4, 5, 7]. Nothing below asserts otherwise, and §8.7 itemizes the provenance of every ingredient in the one place where a cited theorem does load-bearing work.

What is new is the treatment of the relation family as a *variable* rather than as background. The datum studied here is the four-part object

$$\mathfrak{R} = (X, K, G, \mathcal{K}) \longmapsto H^1{}_G(A_K(X), \mathcal{K}),$$

in which the arena and the symmetry sector are held fixed while the declared vocabulary varies over K_{disj}, K_4, K_3, $K_{3,4}$, $K_{\le L}$. The standard model takes the commuting cells and stops; the question here is comparative — which global circulations survive which local language. The distinctive object is the *closure profile* $L \mapsto \varepsilon_{\mathrm{harm}}(K_{\le L})$, and its threshold L_{closure}.

One identification places the comparative picture inside existing literature and we make it explicitly. Grigor'yan, Lin, Muranov and Yau prove that the first path homology of a graph is the cycle space modulo the subspace generated by all 3- and 4-cycles [9, §6]. At $G = 1$ that is exactly our $K_{3,4}$:

$$\varepsilon_{\mathrm{harm}}(K_{3,4} \mid G = 1) \;=\; \dim H_1{}^{\mathrm{path}}(A_K(X); \mathbb{Q}).$$

So the strongest declared vocabulary considered here is not an invention: in the non-equivariant case it computes first path homology of the arena. Measured, this sharpens the dichotomy of §7.3 into standard language. The 66-vertex arena of the §3 witness has cycle rank 235 and rank $d_1(K_{3,4}) = 235$, so its first path homology **vanishes**; every arena of C_n in the tested range has first path homology of dimension **1**. Vocabulary residue is residue the arena does not actually carry; traversal residue is residue it does.

A bounded novelty search was run for the closure profile. Scope: English-language search of arXiv and the open web, October 2026, on cycle-length filtrations of graph configuration spaces, short-cycle fillings in graph braid complexes, cohomology under selective 2-cell attachment, and path-homology filtrations by cycle length, together with backward citation from [9] and [2]. Result: [9] supplies the *fixed* 3-and-4-cycle quotient,

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on a graph rather than an arena, non-equivariantly, and defines no bounded-length family and no vanishing threshold. We found no source defining the parametrized family $K_{\le L}$ or an invariant of L_{closure} type. We therefore classify the closure profile as *apparently new within that search scope*, which is a scoped statement and not a novelty theorem; a single overlooked reference would retract it. The underlying ingredients remain established as above.

1.3 Methodological commitments

Three commitments shaped what is claimed. **First,** every quantity reported here was executed; no number in this paper is quoted from an unrun expression. **Second,** every structural quantity was computed twice, by two engines sharing no logic at the point where an error would be most consequential (§9.1). **Third,** a killed claim is reported, not discarded. Section 9.4 lists seven claims refuted during the investigation, five of them by the author's own controls. Their negative information is part of the result: the reason the earlier search population looked so clean is itself a finding (§3.4).

2. The invariant relation-layer complex

2.1 Token arenas

> **Definition 2.1. (token arena)** Let X be a finite simple graph with vertex set V, and $K \ge 1$. The arena $A_K(X)$ is the graph whose vertices are the K-element subsets $S \subseteq V$, with S adjacent to T whenever $T = (S \setminus \{u\}) \cup \{w\}$ for some $u \in S$, $w \notin S$ with $uw \in E(X)$. Arena vertices are collision-free K-token configurations; arena edges are single-token moves.

Any $G \le \mathrm{Aut}(X)$ acts on $A_K(X)$ by acting on subsets. We write the induced action on arena vertices and on oriented arena edges without further notation.

2.2 Invariant cochains and reversible orbits

Let $C^0{}_G$ be the G-invariant functions on arena vertices and $C^1{}_G$ the G-invariant alternating functions on oriented arena edges. The second space has a feature with no analogue in the non-equivariant setting. If some $g \in G$ carries an oriented arena edge to its own reverse, every invariant alternating cochain must vanish on that edge: the orbit is *reversible* and contributes no coordinate. Hence

$$\dim C^1{}_G \;=\; \#\{\text{oriented edge orbits}\} \;-\; \#\{\text{unoriented edge orbits}\},$$

and $C^1{}_G$ is strictly smaller than a naive orbit count would suggest. This is the single most error-prone step in any implementation, which is why §9.1 attacks it specifically.

The coboundary $d_0 : C^0{}_G \to C^1{}_G$ is the usual difference along an oriented edge. We set

$$\varepsilon \;:=\; \dim C^1{}_G \;-\; \operatorname{rank} d_0\,,$$

the dimension of the space of invariant circulations — the invariant cycle space.

2.3 Declared predicates

> **Definition 2.2. (relation layer)** A *predicate* $\mathcal{K}$ is a set of closed loops in the arena, each declared to bound. It determines $C^2{}_G(\mathcal{K})$, the invariant cochains on the declared cells, and a boundary-evaluation map $d_1 : C^1{}_G \to$

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> $C^2{}_G(\mathcal{K})$ sending a cochain to its circulation around each declared cell. A cell carried to itself with reversed orientation by some group element is reversible and contributes no row.

The predicates used here:

Predicate	Lengths	Cells admitted
Kdisj	4	Commuting squares: two moves along base edges with disjoint closed supports. Four distinct arena vertices.
K_4	4	Every arena 4-cycle. Contains Kdisj.
K_3	3	Every arena 3-cycle. Not a predicate of the original family; see the caveat below.
$K_{3,4}$	3, 4	$K_3 \cup K_4$.
K≤L	3…L	The union of all arena cycle families of length at most L.

Table 2. The declared families. Only the first two belong to the original formulation.

A caveat that is load-bearing throughout. The original definition specifies a 2-cell as a cyclic sequence of *four distinct* arena vertices. Arena 3-cells are therefore *not* 2-cells under that definition, and admitting them extends the complex to cells of mixed size. This is a legitimate generalization but it is a change to the definition, not a third predicate inside the existing family, and we flag it wherever it is used.

2.4 The residue

> **Definition 2.3. (invariant harmonic residue)** For a predicate $\mathcal{K}$, set $\varepsilon_{harm}(\mathcal{K}) := \varepsilon - \text{rank } d_1(\mathcal{K}) = \dim C^1{}_G - \text{rank } d_0 - \text{rank } d_1(\mathcal{K})$. Equivalently $\varepsilon_{harm}(\mathcal{K}) = \dim H^1{}_G(\mathcal{K})$, the dimension of the invariant first cohomology of the declared 2-complex.

Two standard facts are used without proof. **Monotonicity**: $\mathcal{K} \subseteq \mathcal{K}'$ implies $\varepsilon_{harm}(\mathcal{K}') \le \varepsilon_{harm}(\mathcal{K})$, since enlarging the family only adds rows to d_1. **The counting certificate**: if $\varepsilon > \dim C^2{}_G(\mathcal{K})$ then $\varepsilon_{harm}(\mathcal{K}) > 0$, because d_1 cannot have rank exceeding its codomain. The certificate gives a sufficient condition for nonvanishing by dimension count alone; §3.3 notes that it cannot fire on our witness, which is why the witness is informative.

3. Refutation of the two-orbit conjecture

3.1 The conjecture

> **Conjecture 3.1. (two-orbit vanishing; false)** If G acts on X) with exactly two vertex orbits, then $\varepsilon_{harm}(K_4) = 0$.

The motivation was an observed population of 784 cases in which no two-orbit substrate had nonvanishing residue under K_4. Section 3.4 explains why that population could not have contained a counterexample.

3.2 The witness

Let X be the 12-vertex, 30-edge graph whose edge list is given in Appendix A, and take $K = 2$ with $G = \text{Aut}(X)$. Then $|\text{Aut}(X)| = 24$ and X has exactly two vertex orbits, so the hypothesis of Conjecture 3.1 holds. The graph has a clean description:

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- the six vertices of degree 7 induce the octahedron $K(2,2,2)$ — that is, K_6 minus the perfect matching (0,3), (1,4), (2,5);
- the six vertices of degree 3 form an independent set;
- each degree-3 vertex attaches to exactly three octahedron vertices, and in all six cases those three *include an antipodal pair*.

The arena has 66 vertices and 300 edges.

3.3 The certificate

Quantity	Engine A	Engine B	Remark
dim C^1G	22	22	invariant 1-cochains
rank d_0	7	7	coboundaries of invariant 0-cochains
ε	15	15	invariant cycle space
Kdisj cells	291	291	commuting squares (canonical)
rank d_1(Kdisj)	13	13	
ɛharm(Kdisj)	2	2	two surviving directions
K_4 cells	1083	1083	all arena 4-cycles (canonical)
rank $d_1(K_4)$	14	14	one rank gained over Kdisj
ɛharm(K_4)	1	1	nonzero — conjecture refuted

Table 3. The refutation certificate. Engine A builds the invariant cochain space from edge orbits with an explicit reversibility rule; Engine B takes the image of the group-averaging projector and never classifies an orbit (§9.1). The two agree on every intermediate quantity, not merely on the final scalar.

Beyond the rank subtraction, an explicit integral representative h of the surviving class was exhibited, with $d_1(K_4)h = 0$ and orthogonality to im d_0 both verified exactly. The class is realized, not merely counted.

The failure mode is worth naming precisely, because the obvious diagnosis is wrong. This is *not* a shortage of 2-cells: $\varepsilon = 15 = \dim C^2_G(K_{disj})$, so the counting certificate cannot fire. The added relations raise rank only from 13 to 14 while the incoming residue has dimension 2. One direction is killed and one survives. The defect is the *rank deficiency of the added-cell evaluation map*, not a dimensional shortfall — which is what §4 localizes.

3.4 Why the earlier population was uninformative

The 784-case population that motivated Conjecture 3.1 was confined to $\dim H^1_G(K_{disj}) = 1$. In that regime the conjecture's characteristic failure — partial annihilation, $0 < \text{rank } E < r$ where r is the incoming residue dimension — is arithmetically impossible, since $r = 1$ forces rank $E \in \{0, 1\}$ and hence either full closure or no change. The population was a rank-one calibration regime, not a test of the multidimensional statement. The witness has $r = 2$ and rank $E = 1$, the first dimension at which the conjecture can fail by partial annihilation at all. This is not a footnote; it explains why a large clean population carried so little evidential weight.

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4. Localizing the failure: rank access

4.1 The evaluation map

Definition 4.1. (evaluation map) Fix a substrate and the residue $H := H^1_G(K_{disj})$. For a relation family $\mathcal{R}$, let $E_{\mathcal{R}} : H \to C^2_G(\mathcal{R})$ send a class to its vector of circulations around the cells of $\mathcal{R}$. Exact cochains have zero circulation on every closed loop, so this is well defined on classes; the fact was verified computationally for every arena cycle of length 3, 4 and 5 on the witness.

The datum that matters is rank $E_{\mathcal{R}}$, not $|\mathcal{R}|$. This notation is bookkeeping and claims nothing; its value is that it says where to put the measurement. The measurement is what follows.

4.2 A structural classification of arena cycles

At the level of token motion, an arena 4-cycle is one of exactly three things. Classifying by the multiset of base edges traversed:

- **commuting** — two distinct base edges, each used twice in opposite directions; the two tokens move independently. This family is precisely K_{disj}.
- **spectator** — four distinct base edges forming a base 4-cycle, with one token parked throughout and the other walking that 4-cycle.
- **rotation** — four distinct base edges forming a base 4-cycle, with no token parked; the tokens rotate.

Arena 3-cycles split the same way into spectator and rotation; there is no commuting 3-cycle. Both 3-cycle types require a base triangle, a fact we will use decisively in §6.

This classification is itself a cross-check. It recovers K_{disj} by a route that never enumerates pairs of disjoint base edges: of the 1083 arena 4-cycles, exactly 291 are commuting, matching the direct enumeration and its rank 13. Independently, the arena 3-cycle count is predicted as 20·(12 − 3) spectator plus 20 rotation = 200, against 200 enumerated.

4.3 The measured ranks

Relation family $\mathcal{R}$	cells	rank d_1	rank $E\mathcal{R}$
3-cycle, spectator	180	10	2
3-cycle, rotation	20	2	2
3-cycle, all	200	12	2
4-cycle, spectator	528	8	1
4-cycle, commuting (= Kdisj)	291	13	0
4-cycle, rotation	264	7	1
4-cycle, non-disjoint (spectator ∪ rotation)	792	13	1
4-cycle, all	1083	14	1
5-cycle, all	5688	15	2

Table 4. Rank access on the witness's residue H, of dimension 2. The last column is the rank of $E\mathcal{R}$ on H; it is 0 for Kdisj by construction, since H is defined as what Kdisj leaves open.

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The headline comparison is between the second and eighth rows:

1083 square relations supply 1 independent constraint; 20 triangle relations supply 2.

The square vocabulary does not lack cells. It lacks independent access to one of the two residual directions. And the minimal complete reader of this residue is 20 cells — one per base triangle — of a shape the original definition excludes by requiring four distinct arena vertices.

4.4 The shared image line

A sharper fact than the ranks. The spectator and rotation square families are structurally unrelated — one parks a token, the other does not — and each has rank 1. One might expect them to contribute complementary directions that together span H. They do not. In the integral basis of H produced by the computation,

$$\text{im } E_{\text{spectator}} = \text{im } E_{\text{rotation}} = \text{im } E_{\text{all squares}} = \text{span}\{(1, -1/2)\},$$

verified directly by reduced row echelon form on each family's image, not inferred from the coincidence of ranks. The entire square vocabulary, in all three of its structural types, reads one line of a two-dimensional space.

4.5 Detection is not closure

One distinction must be kept, because conflating it produced a wrong value in an earlier draft of this work. Triangles have full rank access to H: rank $E_{\triangle} = 2 = \dim H$. It does not follow that triangles alone close the complex. Measured on the witness:

$$\varepsilon_{\text{harm}}(K_3) = 3, \quad \varepsilon_{\text{harm}}(K_4) = 1, \quad \varepsilon_{\text{harm}}(K_{3,4}) = 0.$$

Triangles alone leave 3 of the 15 invariant circulation directions open; squares alone leave 1; only the union closes everything. The families are not nested — neither $K_3 \subseteq K_4$ nor the reverse — so monotonicity licenses no triangle-versus-square hierarchy. Full rank on one residual quotient is a different statement from closure of the enlarged complex, and the two have different numerical answers here.

5. The omitted-cycle mechanism

5.1 The hypothesis that failed

The surviving class on the witness is detected by an arena 3-cycle. Explicitly, the loop

$$[0,1] \to [1,9] \to [1,2] \to [0,1] \quad \text{circulation } -22587,$$

in which one token sits at vertex 1 while the other walks $0 \to 9 \to 2 \to 0$ around the base triangle $\{0, 2, 9\}$. Since a nonzero circulation on a closed loop certifies non-exactness, and since neither declared predicate contains any 3-cell, the obvious reading was that the residue is the *odd-cycle blind spot* of a 4-cycle-only relation system. That reading made a prediction, and the prediction is false.

5.2 Bipartite arenas that carry residue

> **Proposition 5.1. (the odd-cycle reading is false)** There exist substrates whose arena contains no odd cycle of any length and whose residue under K_4 is nonzero.

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The witnesses are the even cycle graphs. For n even, the arena of C_n at $K = 2$ and $K = 3$ is bipartite — verified by two-colouring — so it contains no odd cycle at all, and yet:

X	K	arena bipartite	odd arena cycles	εharm(K4)	Lclosure	parity
C6	2	yes	none	1	6	even
C8	2	yes	none	1	8	even
C10	2	yes	none	1	10	even
C12	2	yes	none	1	12	even
C6	3	yes	none	1	6	even
C8	3	yes	none	1	8	even

Table 5. Bipartite arenas with nonvanishing residue, under G = Zn. The class is first detected at length n, which is even. No odd cycle exists to detect it.

The residue survives and is first detected at an even length. Parity was incidental. What is operative is whether the cycle class carrying the circulation lies outside the span of the declared boundaries — **omission** from the declared family, at whatever length and whatever parity.

We record this as a methodological point as much as a mathematical one. The odd-cycle reading was not vague; it was precise enough to be tested, and testing it cost one computation. Three successive mechanisms for the §3 witness were proposed in the course of this work and two died this way (§9.4).

6. A residue surviving every short vocabulary

6.1 The collapse on triangle-free bases

> **Lemma 6.1.** If X is triangle-free then $A_K(X)$ contains no 3-cycle, and consequently $K_{3,4} = K_4$ for every K.

Proof. An arena 3-cycle visits three K-sets pairwise differing by one token move. If one token is parked, the mover traverses three base vertices x, y, z with $xy, yz, zx \in E(X)$, a base triangle. If no token is parked, the three moves use base edges bc, ab, ca — again a base triangle. Both cases require a triangle in X, so a triangle-free X admits neither. ■

The lemma was also checked numerically — zero arena 3-cycles on C_n for $n = 5, 6, 7, 9$ at $K = 2$ and $n = 6, 8$ at $K = 3$ — so the collapse is not assumed from the proof alone. Its effect is to reduce the question "is there a substrate with $\varepsilon_{harm}(K_{3,4}) > 0$?" to "is there a triangle-free substrate with $\varepsilon_{harm}(K_4) > 0$?", a considerably easier target.

6.2 On a long cycle, all three predicates coincide

> **Lemma 6.2.** Let $n \geq 5$ and $2 \leq K \leq n - 1$. Then every 2-cell of K_4 in $A_K(C_n)$ is a commuting square, so $K_4 = K_{disj}$ as sets of cells. With Lemma 6.1, $K_{3,4} = K_4 = K_{disj}$.

Proof. Let (A, B, C, D) be an arena 4-cycle on four distinct arena vertices. Over the closed loop each token traces a closed walk in C_n, and the walk lengths sum to 4. A closed walk in C_n is either null-homotopic, hence of even length, or winds, hence of length at least $n \geq 5 > 4$. So no token winds and every token's walk has even length. The partitions of 4 into positive even parts are $\{4\}$ and $\{2, 2\}$.

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In the case {4}, one token makes a closed backtracking walk of length 4 and the rest are parked. Such a walk is $u \to w \to u \to w \to u$ or $u \to w \to y \to w \to u$; the induced arena vertex sequences are (S, T, S, T) and (S, T, U, T), with 2 and 3 distinct vertices respectively. Neither has four distinct arena vertices, so this case contributes no 2-cell.

In the case {2, 2}, two tokens each traverse one base edge and return. Writing the moves as $u \leftrightarrow w$ and $v \leftrightarrow x$, four distinct arena vertices force the two moves to interleave, giving the square $\{u,v\} \to \{w,v\} \to \{w,x\} \to \{u,x\}$. The base edges uw and vx must have disjoint closed supports, since a shared vertex would have to be simultaneously occupied and free. That is precisely a commuting square. ■

The bound $n \geq 5$ is sharp and the reason is visible in the proof: for $n = 4$ a token can wind, because C_4 *is* a base 4-cycle, and spectator and rotation squares then exist. This was confirmed computationally — $K_4 = K_{disj}$ on every tested row with $n \geq 5$ across all admissible K, and *not* on C_4, where 6 arena 4-cycles meet only 2 commuting squares.

6.3 The main theorem

> **Theorem 6.3.** Let $n \geq 5$ and $2 \leq K \leq n - 1$. Then $\varepsilon_{harm}(K_{3,4}) = 1$ for $G = 1$ and for $G = \mathbb{Z}_n$, and $\varepsilon_{harm}(K_{3,4}) = 0$ for $G = D_{2n}$. In particular $\varepsilon_{harm}(K_{3,4}) > 0$ occurs on an infinite family.

Proof. By Lemmas 6.1 and 6.2 the three predicates coincide, so it suffices to treat K_{disj}. For $G = 1$, §8.1 gives $\varepsilon_{harm}(K_{disj}) = \dim H^1(UD^K(C_n); \mathbb{Q})$ — an identity, since the declared cells are the 2-skeleton of the discrete configuration space and H^1 does not see higher cells (§8.2). Since $n \geq K + 1$, Corollary 8.2 verifies Abrams' hypothesis, so $UD^K(C_n) \simeq \mathrm{UConf}_K(C_n) \simeq \mathrm{UConf}_K(S^1) \simeq S^1$, giving dimension 1.

For a finite group G and $\mathbb{Q}$ coefficients, averaging over G is a projector commuting with the differentials, so taking invariants is exact and $H(C^\bullet)^G \cong H(C^\bullet{}_G)$; hence ε_{harm} computes $\dim (H^1)^G$. By §8.6 every rotation acts as +1 and every reflection as −1 on the one-dimensional H^1, so the $\mathbb{Z}_n$-invariants are all of it and the D_{2n}-invariants are zero. ■

Two remarks on what this is and is not. It *is* a proof for the stated infinite family, resting on Abrams' theorem as a cited input with its hypothesis verified, not on the tabulated computations; the computations below are a check on the implementation rather than the evidence for the statement. It is *not* a proof of the closure-depth formula of §7, which remains a tested pattern — see the caveat there.

n	dim C¹G	rank d_0	ε	K₄ cells	rank d_1	εharm (Zn)	εharm (D2n)
5	3	1	2	5	1	1	0
6	4	2	2	9	1	1	0
7	5	2	3	14	2	1	0
8	6	3	3	20	2	1	0
9	7	3	4	27	3	1	0
10	8	4	4	35	3	1	0
11	9	4	5	44	4	1	0
12	10	5	5	54	4	1	0

Table 6. The cycle family at K = 2 under the rotation subgroup, with the full dihedral column for contrast — both as predicted by Theorem 6.3. Each row was also computed under the trivial group, giving 1. In total 42 rows were measured across n = 5…12 and K = 2, 3, of which 28 are positive: exactly the trivial-group and Zn rows.

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6.4 In path-homology terms

Because $K_{3,4}$ at trivial group computes first path homology of the arena (§1.3), Theorem 6.3 can be restated without reference to this paper's machinery: for $n \geq 5$ and $2 \leq K \leq n - 1$, the token arena $A_K(C_n)$ has first path homology of dimension 1 over $\mathbb{Q}$. By contrast the 66-vertex arena of the §3 witness has cycle rank 235 and rank $d_1(K_{3,4}) = 235$, so its first path homology vanishes. The two regimes of §7.3 are then distinguishable in standard language: vocabulary residue is residue the arena does not carry, and traversal residue is residue it does.

6.3 The group is a second, independent dial

The dihedral column is not noise. The same substrate with the same declared cells yields residue 1 under $\mathbb{Z}_n$ and 0 under D_{2n}). Changing the cell vocabulary and changing the symmetry group are mathematically distinct operations: the first decides which circulations become boundaries, the second decides which classes survive invariance. Section 8.6 computes exactly why the reflections destroy this class, which converts the observation into a representation-theoretic statement.

7. Closure depth

7.1 Definition

> **Definition 7.1. (closure depth)** For a substrate (X, K, G), let L_{closure} be the least $L \geq 3$ with $\varepsilon_{\text{harm}}(K_{\leq L}) = 0$, where $K_{\leq L}$ is the union of all arena cycle families of length at most L; and ∞ if no such L exists.

Monotonicity makes $\varepsilon_{\text{harm}}(K_{\leq L})$ non-increasing in L, so L_{closure} is well defined as a threshold. It measures not how complicated the substrate is, but *the shortest relation depth that separates every surviving invariant circulation class*. Because the arena is finite, admitting all cycles up to its circumference closes everything, so L_{closure} is finite for any single finite substrate; the content is how it behaves across a family.

7.2 The ladder

X	K	L≤3	L≤4	L≤5	L≤6	L≤7	L≤8	L≤9	Lclosure
C5	2	2	1	0	—	—	—	—	5
C6	2	2	1	1	0	—	—	—	6
C7	2	3	1	1	1	0	—	—	7
C8	2	3	1	1	1	1	0	—	8
C9	2	4	1	1	1	1	1	0	9
C10	2	4	1	1	1	1	1	1	10
C6	3	3	1	1	0	—	—	—	6
C7	3	6	1	1	1	0	—	—	7
witness	2	3	0	—	—	—	—	—	4

Table 7. εharm(K≤L) as the declared cycle length grows, under G = Zn for the cycle rows and G = Aut(X) for the witness. Dashes are lengths at or beyond closure. The cycle rows are a step function: flat at 1 from L = 4 through L = n − 1, then 0 at L = n.

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Observation 7.2. (computational; range as stated) $L_{closure}(C_n, K) = n$ on every row of Table 7, and $L_{closure} = 4$ for the §3 witness.

This is a tested pattern, not a theorem, and the distinction matters for what may be concluded. Theorem 6.3 proves that the residue is nonzero under $K_{3,4}$ for the whole infinite family, which already gives $L_{closure} > 4$ for every $n \geq 5$. What is *not* proved is the exact value: that would require showing that every arena loop of length strictly less than n lies in the span of the declared boundaries of $K_{\leq n-1}$, for all n and all admissible K — a statement about the span of short cycles in the arena that Abrams' theorem does not supply. Accordingly:

Corollary 7.3. (from Theorem 6.3, not computational) For every $n \geq 5$ and $2 \leq K \leq n - 1$, $L_{closure}(C_n, K) > 4$ under $G = 1$ and $G = \mathbb{Z}_n$). Hence no uniform L closes the cycle family — the supremum of closure depth over it is infinite — *provided* the measured growth $L_{closure} = n$ is granted beyond the tested range; the unconditional consequence of Theorem 6.3 alone is that no $L \leq 4$ suffices for any member.

The witness row deserves a note, since an earlier draft of this work recorded $L_{closure} = 3$ for it. That was wrong, by exactly the conflation of §4.5: triangles *detect* the full residue but do not *close* the complex, leaving 3 of 15 directions open. The union up to length 4 is the first that closes everything.

7.3 Two populated regimes

	bounded regime	substrate-tracking regime
representative	§3 witness, 12 vertices	Cn, every n ≥ 5
Lclosure	4	n
behaviour across the family	bounded	tracks the substrate; sup = ∞
obstruction	a declared family with incomplete rank access to the residue	a global traversal class unreadable below length n
repair	admit the omitted bounded-length family	admit a substrate-scale relation
uniform finite vocabulary	succeeds	cannot succeed uniformly in n
name used here	vocabulary residue	traversal residue

Table 8. The two observed regimes. This is a description of two populated cases, not a classification theorem: nothing here shows every residue falls into one of them.

The distinction has a clean reading. For vocabulary residue the declared language was missing a finitely describable relation family; admit it and the class evaporates while the substrate never moved. For traversal residue the class follows a global route through the substrate, and the relation needed to close it is not uniformly bounded in length. The first says the apparent flatness was apparent; the second says that at any bounded reading depth there is genuinely no shorter address.

8. Identification of the traversal class

On C_n at $K = 2$ the shortest nonzero-circulation loop has length n and returns the configuration to its starting set with the two token labels exchanged — verified by tracking labels around the loop for $n = 5, 6, 7, 8$. That suggests the half-twist of an unordered two-point configuration space. A suggestion is not an identification, so this section builds the identification in four separable steps and marks what is cited, what is standard, and what is computed here.

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8.1 A definitional identity at trivial group

When $G = 1$ there are no reversible orbits and no averaging; $C^1{}_G$ is the full oriented-edge cochain space. Then

$$\varepsilon_{harm} = \dim C^1 - \operatorname{rank} d_0 - \operatorname{rank} d_1 = \dim \ker d_1 - \dim \operatorname{im} d_0 = \dim H^1(2\text{-complex}; \mathbb{Q}),$$

which is simplicial first cohomology of the declared 2-complex over $\mathbb{Q}$, by definition rather than by analogy. This is the step that makes the rest checkable.

8.2 The declared cells are the 2-skeleton of the discrete configuration space

Abrams' *discretized configuration space* $UD^K(X)$ is the cube complex whose d-cells are given by d base edges with pairwise disjoint closed supports, one token on each and free to sit at either end, with the remaining $K - d$ tokens parked on vertices disjoint from all of them. Comparing against Definition 2.1 and the first row of Table 2:

- 0-cells are collision-free K-token configurations — the arena vertices;
- 1-cells are single-token moves along a base edge — the arena edges;
- 2-cells are pairs of moves on disjoint closed supports — precisely K_{disj}.

So K_{disj} supplies the cells of $UD^K(X)$ *through dimension two* and no higher. The qualification is necessary and is not a weakness for our purpose: first cohomology of a CW complex is computed by the cochain complex $C^0 \to C^1 \to C^2$, so cells of dimension 3 and above do not affect H^1). Therefore

$$\varepsilon_{harm}(K_{disj} \mid G = 1) \;=\; \dim H^1(UD^K(X); \mathbb{Q}).$$

We say "2-skeleton" rather than "the complex" throughout, since the higher commuting cubes are not built.

8.3 Verifying the cell structure independently

The identification in §8.2 is a claim about our implementation, so it is testable against a closed form. For C_n, the number of d-cubes is $m_d(C_n) \cdot C(n - 2d, K - d)$, where m_d counts d-edge matchings of the cycle. We checked the matching counts against brute force for all $n \le 12$ and all d (exact agreement), then compared cube counts with the enumeration and formed the alternating sum.

X	K	#0	#1	#2	#3	#4	χ (full)	χ(S¹)	χ (2-skel)
C5	2	10	15	5	—	—	0	0	0
C6	3	20	36	18	2	—	0	0	2
C7	3	35	70	42	7	—	0	0	7
C8	4	70	160	120	32	2	0	0	30
C9	4	126	315	270	90	9	0	0	81
C10	3	120	280	210	50	—	0	0	50

Table 9. Cube counts from the closed form. Dimensions 0, 1 and 2 matched the enumeration on every one of the 33 admissible rows. The full alternating sum is 0 on every row, agreeing with $\chi(S^1) = 0$. The 2-skeleton's characteristic is in general nonzero and equals exactly the alternating sum of the omitted higher cubes — for C8 at $K = 4$, $32 - 2 = 30$ — which quantifies the §8.2 qualification instead of merely conceding it.

Two conclusions. The cells we build are the cells of UD^K; and the complex they sit inside has the Euler characteristic of a circle on every admissible row, by a route independent of any rank computation.

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8.4 Abrams' theorem and its hypothesis for the cycle

> **Theorem 8.1. (Abrams; cited, not proved here)** Let G be a graph with at least n vertices. Then $C^n(G)$ deformation retracts onto $D^n(G)$ if (A′) each path connecting distinct essential vertices of G has length at least $n + 1$, and (B′) each homotopically essential path connecting a vertex to itself has length at least $n + 1$.

We use the statement as given by Prue and Scrimshaw [2], who also improve (A′) from $n + 1$ to $n - 1$ by discrete Morse theory while leaving (B′) unchanged. The descent of the retraction from ordered to unordered configurations is the standard free-quotient argument, the symmetric group acting freely on both sides.

> **Corollary 8.2.** For $X = C_n$, conditions (A′) and (B′) reduce to the single condition $n \geq K + 1$.

Proof. Every vertex of C_n has degree 2, so there are no essential vertices and (A′) is vacuous — the Prue–Scrimshaw improvement is therefore irrelevant in this case. The only homotopically essential loop is the full cycle, of length n; (B′) requires $n \geq K + 1$. ■

Combined with the classical fact that $\mathrm{UConf}_K(S^1) \simeq S^1$, so that $\dim H^1(\mathrm{UConf}_K(S^1); \mathbb{Q}) = 1$ for every K, the chain is

$$UD^K(C_n) \;\simeq\; \mathrm{UConf}_K(C_n) \;\simeq\; \mathrm{UConf}_K(S^1) \;\simeq\; S^1 \quad (n \geq K + 1).$$

8.5 The measured range and the first excluded row

A hypothesis with a boundary makes the identification a test rather than a decoration. Measured: $\varepsilon_{\mathrm{harm}}(K_{\mathrm{disj}} \mid G = 1) = 1$ on all **33** admissible rows, $n = 3 \ldots 10$ and $2 \leq K \leq n - 1$; and 0 on all **8** rows with $K = n$, the first excluded case, where every vertex is occupied, no token can move, and the discrete model is a single point — while $\mathrm{UConf}_n(S^1)$ is still $\simeq S^1$.

The logic of that boundary row must be stated carefully. A failed hypothesis does not make a theorem's conclusion false in general; the theorem simply asserts nothing there. The correct statement is: *the discrete computation agrees with the configuration-space prediction throughout the verified Abrams range, and differs from it at the first excluded row $K = n$, where the loop-length condition fails.* It happens that this excluded case is one where the conclusion is in fact false, which is what makes the row a useful calibration: it shows the hypothesis is doing work in our range rather than being cited decoratively.

8.6 The equivariant degree

Section 6.3 observed that the full dihedral group destroys the class. That observation can be upgraded to a computation. Let h span the rank-one cohomology at trivial group. For $g \in \mathrm{Aut}(C_n)$, the class $[g^*h]$ is a multiple of $[h]$, so there is a unique λ_g with

$$g^*h \;=\; \lambda_g\, h \;+\; d_0 f_g \quad \text{for some } f_g \in C^0.$$

Solving this as an exact linear system in the unknowns (λ_g, f_g) — not by testing candidate values — gives, on 20 rows spanning $n = 5 \ldots 9$ and $K = 2, 3$, every system consistent and

$$\boldsymbol{\lambda_g = +1 \text{ for every rotation}, \quad \lambda_g = -1 \text{ for every reflection.}}$$

> **Corollary 8.3.** The rank-one cohomology carries the trivial character under $\mathbb{Z}_n$ and the sign character under D_{2n}). Hence $(H^1)^{\mathbb{Z}_n} \cong \mathbb{Q}$ while $(H^1)^{D_{2n}} = 0$.

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This is the dihedral column of Table 6, derived. The reflection reverses the traversal orientation, identifying the class with its negative; the invariant subspace of a one-dimensional space carrying a −1 action is zero. Rotations are homotopic to the identity on a circle and act trivially, which is the other column.

8.7 Scope: what is cited, standard, computed, and outstanding

Ingredient	Status	Note
UDK ≃ UConfK under (A′),(B′)	cited	Abrams' theorem [1], via the statement in [2]; hypothesis verified for Cn in Corollary 8.2, not reproved
descent to unordered configurations	standard	free action of the symmetric group on both sides
UConfK(S^1) ≃ S^1	standard	classical
H^1 depends only on cells of dim ≤ 2	standard	cellular cochain complex
ε$harm$ = dim H^1 at G = 1	computed / definitional	§8.1
cube counts and χ = 0 on 33 rows	computed	§8.3, closed form vs enumeration
rank 1 on 33 admissible rows, 0 on 8 boundary rows	computed	§8.5
λ = +1 rotations, −1 reflections, 20 rows	computed	§8.6, exact linear solve
explicit chain map from this implementation to a standard UDK model	outstanding	not written down
chain-level identification of h with a canonical generator	outstanding	not written down

Table 10. Provenance of each ingredient in the identification. The two outstanding rows limit how strongly the specific representative may be described; they do not weaken §8.1 or §8.5.

9. Methods

9.1 Two engines

Every structural quantity was computed by two implementations that differ precisely where an error would be most consequential: the construction of C^1_G.

Engine A follows the formulation directly. It enumerates orbits of oriented arena edges, applies an explicit reversibility test — does any group element carry this oriented edge to its reverse? — discards reversible orbits, and takes the surviving orbit representatives as a basis. Cells are canonicalized by minimizing over rotations and reversed rotations, with an orientation sign.

Engine B shares none of that logic. It builds the signed action of each group element on the full oriented-edge coordinate space, forms the group-averaging projector $P = \Sigma_g T_g$, and takes the image of P as the invariant subspace. Reversible directions are annihilated by P as a consequence of the representation; they are never detected, classified, or discarded. No orbit is enumerated and no canonical form is computed anywhere in Engine B.

If Engine A's reversibility bookkeeping were wrong, the two would disagree. On 24 rows spanning the cycle family at K = 2, 3 under the trivial group and under $\mathbb{Z}_n$, they agree on all eight reported quantities — arena

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vertices, arena edges, dim C^1_G, rank d_0, ε, cell count, rank d_1, and ε_{harm} — with zero disagreements. They also agree on every intermediate quantity of the §3 refutation certificate (Table 3), which matters because two implementations can converge accidentally on a derived scalar while differing on its inputs.

Both engines are the author's, so this is reproduction under an independent construction, not independent authorship. We state that limit rather than imply otherwise.

9.2 Exact arithmetic

All linear algebra is over $\mathbb{Q}$ with Python fractions.Fraction, by fraction-free Gaussian elimination. No floating point appears anywhere in the computation, so no rank is reported subject to a tolerance. Ranks were additionally cross-checked modulo two large primes during development. The surviving cohomology representatives are reported as primitive integer vectors after clearing denominators and dividing by the content.

9.3 Cross-checks that passed

Check	What was compared	§
Independent recovery of Kdisj	Classifying all 1083 arena 4-cycles by move-multiset yields exactly 291 commuting squares, rank $d_1 = 13$ — matching the direct enumeration by a route that never pairs base edges.	§4.2
Arena 3-cycle count	Predicted 20·(12−3) spectator + 20 rotation = 200; enumerated 200.	§4.2
Exactness of im d_0	Every arena cycle of length 3, 4, 5 has zero circulation on every coboundary.	§4.1
Lemma 6.1 numerically	Zero arena 3-cycles on triangle-free bases, so the $K_{3,4} = K_4$ collapse is not assumed from the proof alone.	§6.1
Matching counts	Closed form vs brute force for all $n \le 12$ and all d: exact agreement.	§8.3
Cube counts	Closed form vs enumeration at dimensions 0, 1, 2 on all 33 admissible rows.	§8.3
Euler characteristic	Full alternating sum = 0 = $\chi(S^1)$ on all 33 rows; 2-skeleton deficit equals the alternating sum of omitted higher cubes.	§8.3
Connectivity	rank d_0 = (arena vertices) − 1 on every connected row, confirming $H^0 = \mathbb{Q}$.	§8.1
Engine agreement	24 rows × 8 quantities, 0 disagreements; plus the full §3 certificate.	§9.1
Equivariant consistency	All 20 linear systems $g^*h = \lambda h + d_0 f$ solvable; $\lambda = \pm 1$ as predicted.	§8.6

Table 11. The cross-checks. Several are redundant with the main computation by design: a check that cannot fail is not a check, so each row above compares two routes that could in principle disagree.

9.4 Claims refuted during the investigation

Seven claims entertained in the course of this work were killed. They are recorded because a killed claim is a constraint, and because two of the three mechanisms proposed for the §3 witness died by their own predictions — which is the reason the surviving account in §4 is about rank rather than shape.

Claim	How it died	§
Two vertex orbits force $\varepsilon harm(K_4) = 0$	exact 12-vertex witness, two engines	§3
The residue is an odd-cycle blind spot of a 4-cycle-only predicate	bipartite arenas, no odd cycle of any length, residue 1	§5.2
Adding arena 3-cells annihilates the residue universally	Cn, triangle-free, $\varepsilon harm(K_{3,4}) = 1$	§6.2

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Claim	How it died	§
Lclosure = 3 for the §3 witness	conflated detection with closure; triangles leave 3 of 15 open, so it is 4	§7.2
Cell shape of the added 4-cycles is the discriminator	three square types, same image line	§4.4
Orbit size of the added cells is the discriminator	own control, refuted	—
gcd of the primitive survivor equals dim C^1G × |Aut|	own control: held in 8 of 120 cases, all from one graph where 112 = 14 × 8 coincidentally	—

Table 12. Refuted claims. Five of the seven were killed by the author's own controls rather than by external review.

One further correction is methodological rather than mathematical. An earlier write-up described the §8.5 boundary row as the Abrams identification "failing at its boundary." That misstates the logic of a hypothesis: a theorem asserts nothing where its hypothesis fails, so nothing of it fails there. The wording in §8.5 is the corrected form.

10. Open problems

10.1 The bounded-closure problem

> **Problem 10.1. (bounded closure)** Give structural conditions on a family of substrates (X, K, G) under which there exists a uniform L with $\varepsilon_{harm}(K_{\leq L}) = 0$ throughout the family — equivalently, under which $L_{closure}$ is uniformly bounded.

Theorem 6.3 and Observation 7.2 together show the obstruction is real: closure depth exceeds 4 for every member of an infinite family and is measured to grow as n, so any bounded-closure theorem must exclude, quotient out, or otherwise account for the configuration-space traversal component. A natural first guess is that simple connectivity of the arena suffices. One care point says it is not immediate: $H_1(A_K(X); \mathbb{Q}) = 0$ does not by itself give $H^1_G(K_{\leq L}) = 0$ at bounded L, because a declared system need not span all null-homologous cycles at a uniformly bounded local scale. Whether the vocabulary regime admits a uniform bound is the live question, and it is a question about token arenas in general rather than about the witness of §3.

10.2 Secondary questions

(i) **Is the dichotomy exhaustive?** Table 8 describes two populated regimes. Nothing here shows every residue is of one of these kinds, and a substrate with $L_{closure}$ growing more slowly than the substrate but unboundedly would be a third behaviour.

(ii) **A chain-level map.** Writing an explicit chain map from the arena implementation to a standard UD^K model, and identifying the computed representative with a canonical generator, would close the two outstanding rows of Table 10.

(iii) **Characterize rank access.** On the §3 witness the whole square vocabulary reads one line. What property of a substrate determines the rank of E for the square family? The three discriminators tried here — cell shape, orbit size, and a gcd invariant of the primitive survivor — were all refuted (Table 12), and all three were searching inside the added cells. The discriminator is presumably a relation between the substrate's cycle space and the span of the declared boundaries.

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(iv) **Mixed-size complexes.** The caveat of §2.3 stands: admitting arena 3-cells changes the definition of a 2-cell. A formulation in which cells of mixed boundary length are first-class, with its own monotonicity and counting certificates, would make $K_{\leq L}$ a native object rather than an extension.

11. Scope of claims

Stated plainly, to prevent the computational results being read as more than they are.

Established. The two-orbit conjecture is false, by an exact witness reproduced under two constructions. The odd-cycle mechanism is false, by bipartite witnesses. $\varepsilon_{harm}(K_{3,4}) > 0$ on an infinite family — Theorem 6.3, proved rather than observed, using Abrams' theorem as a cited input with its hypothesis verified. $L_{closure} = n$ for the measured cycle rows and 4 for the witness, with $L_{closure} > 4$ proved for the whole family. On the witness's two-dimensional residue the square vocabulary has rank 1 with a single shared image line and the triangle families have rank 2. At trivial group the computed quantity is dim H^1 of the declared 2-complex over $\mathbb{Q}$, the declared cells are the 2-skeleton of UD^K, the cube counts and Euler characteristics agree with closed form, Abrams' hypothesis for C_n is $n \geq K + 1$, the measured rank is 1 on all 33 admissible rows and 0 on all 8 excluded rows, and the equivariant degree is +1 on rotations and −1 on reflections.

Not established. That every invariant harmonic residue belongs to one of the two regimes of Table 8. That vocabulary residue admits a uniform closure bound. The exact value of $L_{closure}$ outside the tested rows: Corollary 7.3 proves only that it exceeds 4. An explicit chain map from this implementation to a standard UD^K model, or a chain-level identification of the computed representative with a canonical configuration-space generator. Abrams' theorem, which is cited and not reproved. Any statement for substrates, groups, or values of K outside the ranges tabulated above: every computational claim here is a finite computation over a stated range, and none of it is a proof of a general theorem.

Not claimed at all. Nothing physical. ε_{harm} is the dimension of an invariant first cohomology under a declared predicate. The words "reader", "residue", "access" and "vocabulary" are names for rank computations, chosen because they are shorter than the computations; they carry no claim about physical observers, measurement, or information in any thermodynamic sense.

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[8] Kulik, D. *The Relation Layer: invariant cochain complexes over token-configuration arenas.* QuHarmonics Research Group, 2026. (The source of Conjecture 3.1 and of Definitions 2.1–2.3.)

[9] Grigor'yan, A., Lin, Y., Muranov, Y. and Yau, S.-T. *Homologies of path complexes and digraphs.* arXiv:1207.2834. (First path homology = cycle space modulo 3- and 4-cycles, their §6; this is our $K_{3,4}$ at G = 1. The closest literature analogue, and the basis of §1.3 and §6.4.)

[10] Abrams' stabilization theorem for no-k-equal configuration spaces on graphs. arXiv:2407.07854.

Appendix A. The witness

The 12-vertex graph X of §3, as an edge list on vertices 0–11:

(0,1) (0,2) (0,4) (0,5) (0,6) (0,7) (0,9) (1,2) (1,3) (1,5)
(1,7) (1,8) (1,10) (2,3) (2,4) (2,8) (2,9) (2,11) (3,4) (3,5)
(3,6) (3,9) (3,10) (4,5) (4,7) (4,10) (4,11) (5,6) (5,8) (5,11)

Invariants: 12 vertices, 30 edges, girth 3, 20 triangles, $|\mathrm{Aut}(X)| = 24$, exactly two vertex orbits (the six degree-7 vertices and the six degree-3 vertices). Degree-7 part: the octahedron $K(2,2,2)$ with antipodal pairs (0,3), (1,4), (2,5). Degree-3 attachments, each containing an antipodal pair:

$6 \to \{0,3,5\}$ $7 \to \{0,1,4\}$ $8 \to \{1,2,5\}$ $9 \to \{0,2,3\}$ $10 \to \{1,3,4\}$ $11 \to \{2,4,5\}$

At $K = 2$ the arena has 66 vertices and 300 edges. The arena 3-cycle detecting the surviving K_4-harmonic class is $[0,1] \to [1,9] \to [1,2] \to [0,1]$, circulation −22587, encoding the base triangle {0, 2, 9}.

Appendix B. Reproduction

All results were produced by a single self-contained Jupyter notebook, two_species_residue.ipynb, of 18 cells, executed end to end from a clean kernel with zero errors. It imports nothing outside the Python standard library: itertools, fractions, math, collections. Both engines of §9.1 are defined in its first code cell.

Section correspondence: §4 is notebook §1; §6 is notebook §2; §7.2 is notebook §3; §8 is notebook §4; §5 is notebook §5; §9.1 is notebook §6; Tables 7 and 8 are notebook §7. The Euler-characteristic and matching-count verification of §8.3 is a separate script, euler.py, of the same provenance.

Runtime is a few minutes on one core for everything except the length-5 arena cycle enumeration on the witness (5688 cells) and the closure ladders, which dominate. Every table in this paper is a transcription of live output; no value was computed by hand.

Appendix C. Full predicate table for the witness

Predicate	cells	rank d_1	εharm	remark
Kdisj	291	13	2	
K3 (triangles only)	200	12	3	not a predicate of the original family

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Predicate	cells	rank d_1	εharm	remark
K_4 (all arena 4-cycles)	1083	14	1	refutes Conjecture 3.1
K_5 (all arena 5-cycles)	5688	15	0	
$K_{3,4}$	1283	15	0	
$K \leq 5$	6971	15	0	

Table 13. Every predicate measured on the §3 witness at K = 2, with ε = 15 and dim C^1G = 22, rank d_0 = 7 throughout. The non-nesting of K_3 and K_4 is visible in rows 2 and 3: neither rank dominates the other, and only their union reaches 15.

Note that K_5 alone also reaches rank 15. Closure at length 5 without length 4 is possible here; $L_{\text{closure}} = 4$ because the definition takes the union of all lengths up to L, and the union through 4 already closes.

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